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**Master of Computer Applications
Semester – III
Data Analytics for Business
Experiment No.: 06
Title: Creating HR analytics dashboards**

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Submitted To:

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**Experiment No.: 06
Creating HR analytics dashboards**

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Problem Statement

You are an HR Data Analyst in a multinational organization. The HR department wants insights into employee demographics, hiring trends, attrition, and workforce distribution. Develop an HR analytics dashboard to support workforce planning and employee management.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained Hr-Analytics Attrition data from Kaggle datasets.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported all datasets using the **Get Data** option.
- Loaded CSV file into Power BI.

Step 3: Data Cleaning and Transformation

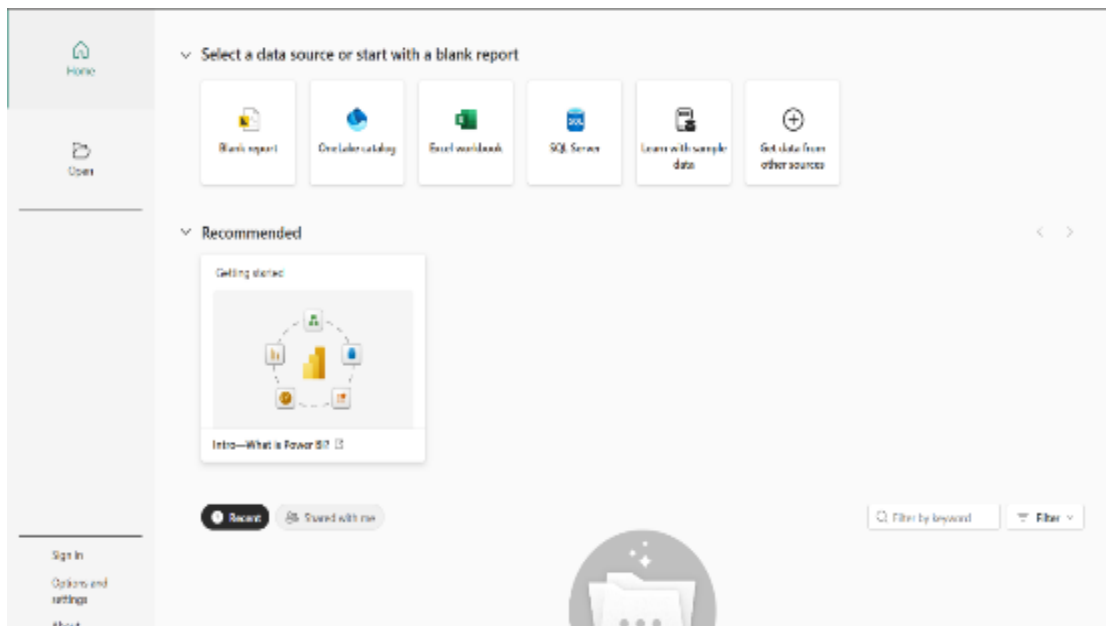
- Removed duplicate records.
- Handled missing values.
- Converted data types where necessary

Step 4: Dashboard Development

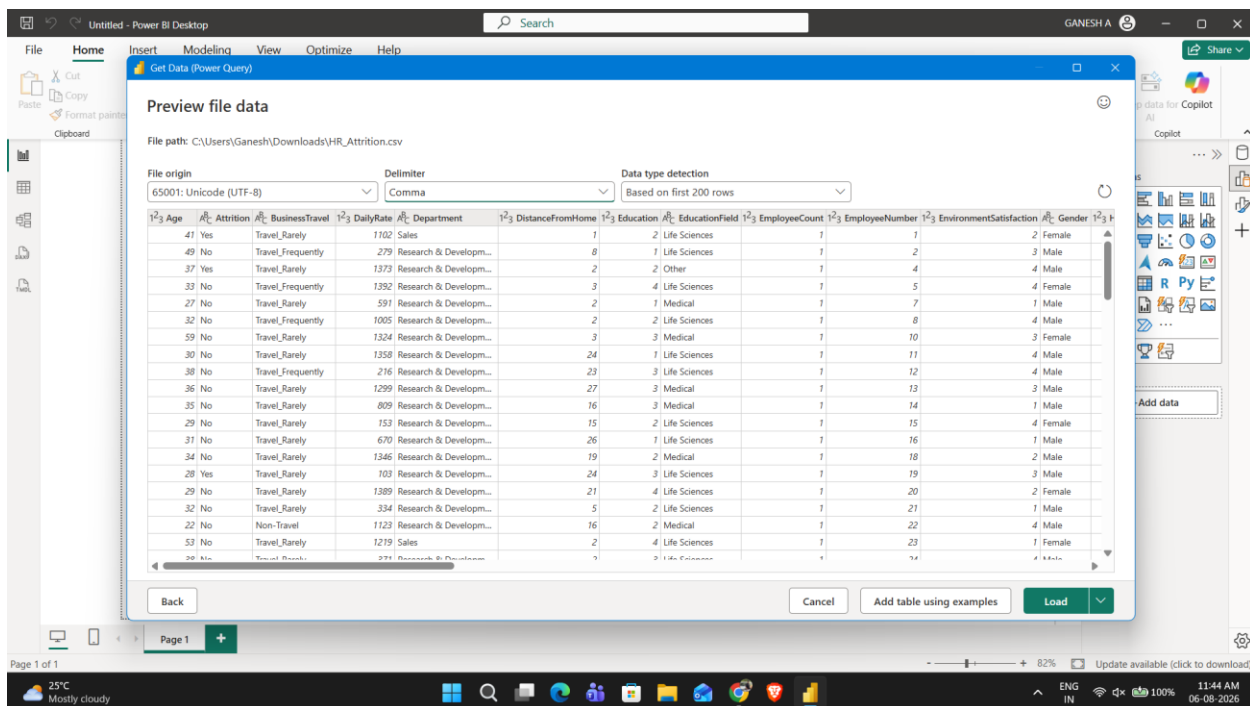
- Created KPI Cards to display Total Headcount, Active Employees, Average Monthly Income, Attrition Count, and Female Ratio
- Developed a Clustered Bar Chart to analyze Employee Count and Attrition across different Departments and Job Roles.
- Developed a Stacked Column Chart to evaluate the impact of OverTime on Attrition rates.
- Created a Donut Chart to analyze the overall proportion of Retained vs. Attrited employees.
- Created a Gauge Chart to track average Job Satisfaction levels against performance benchmarks.
- Developed a Line Chart to visualize workforce distribution and turnover trends across employee Age progression.
- Developed a Bar Chart to analyze Business Travel frequency and its relation to workforce capacity.

Output

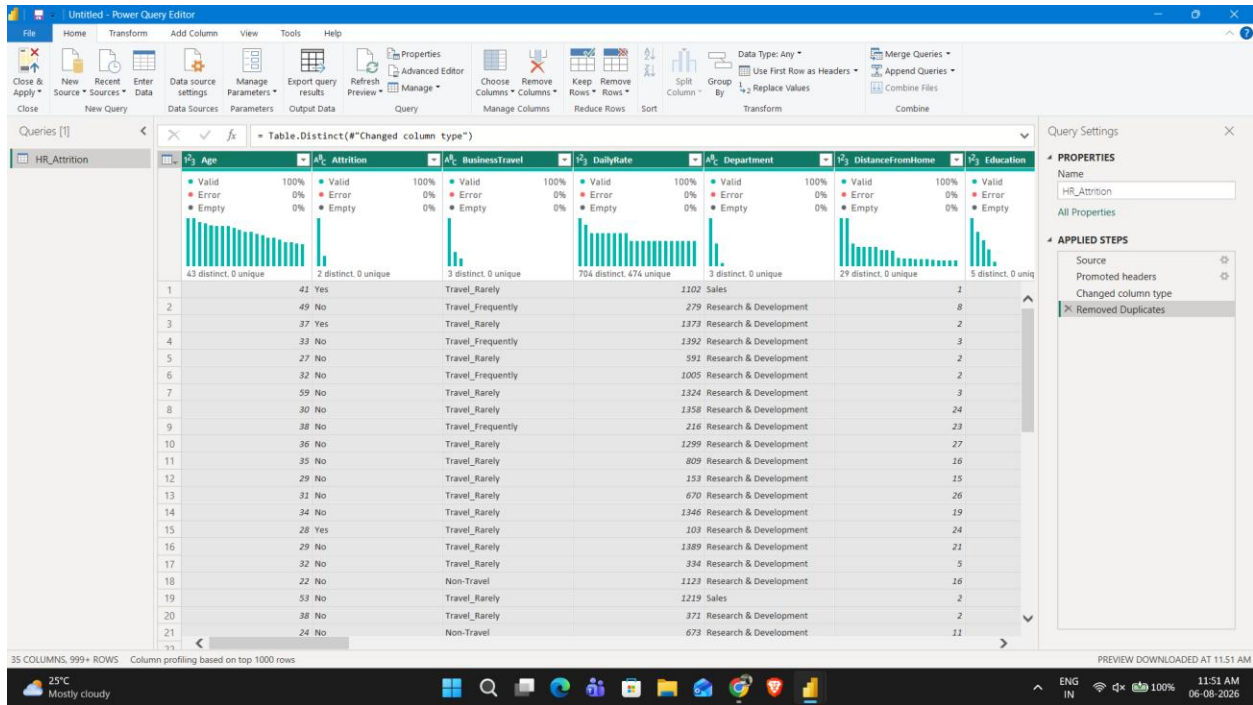
- Opened Power BI file.



- Loaded the dataset.

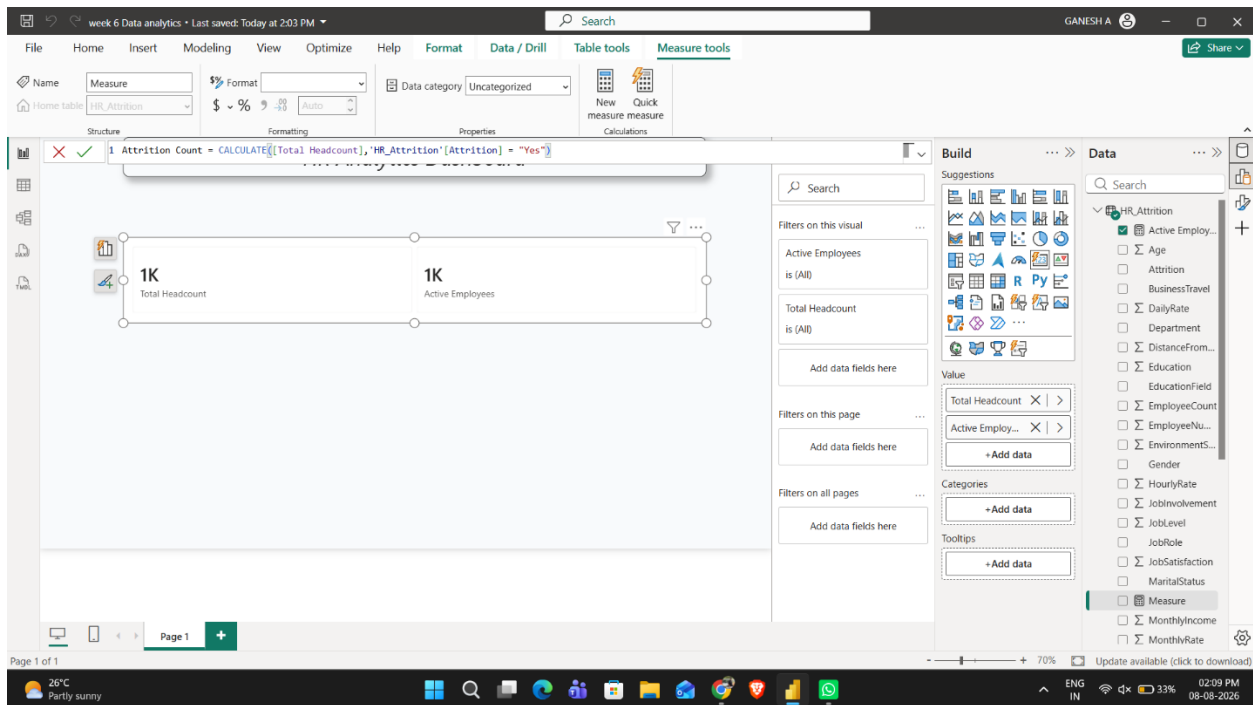


- Transformed data And applied Cleaning where ever Necessary.

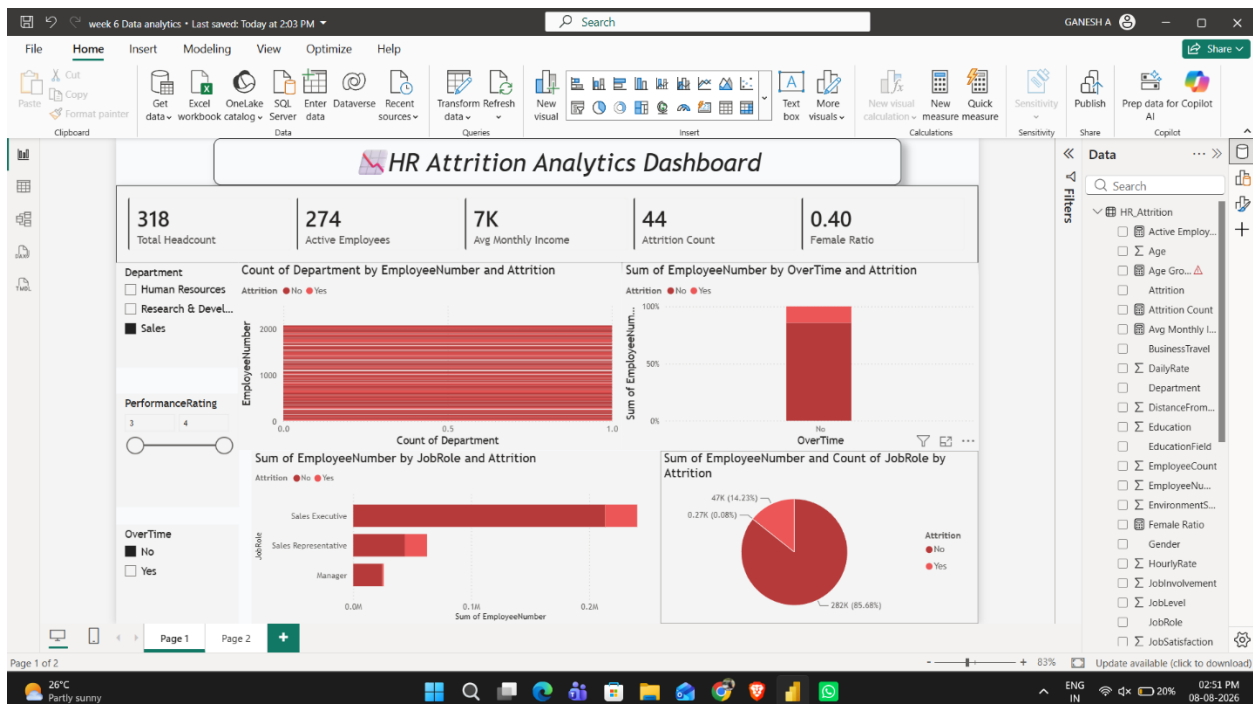


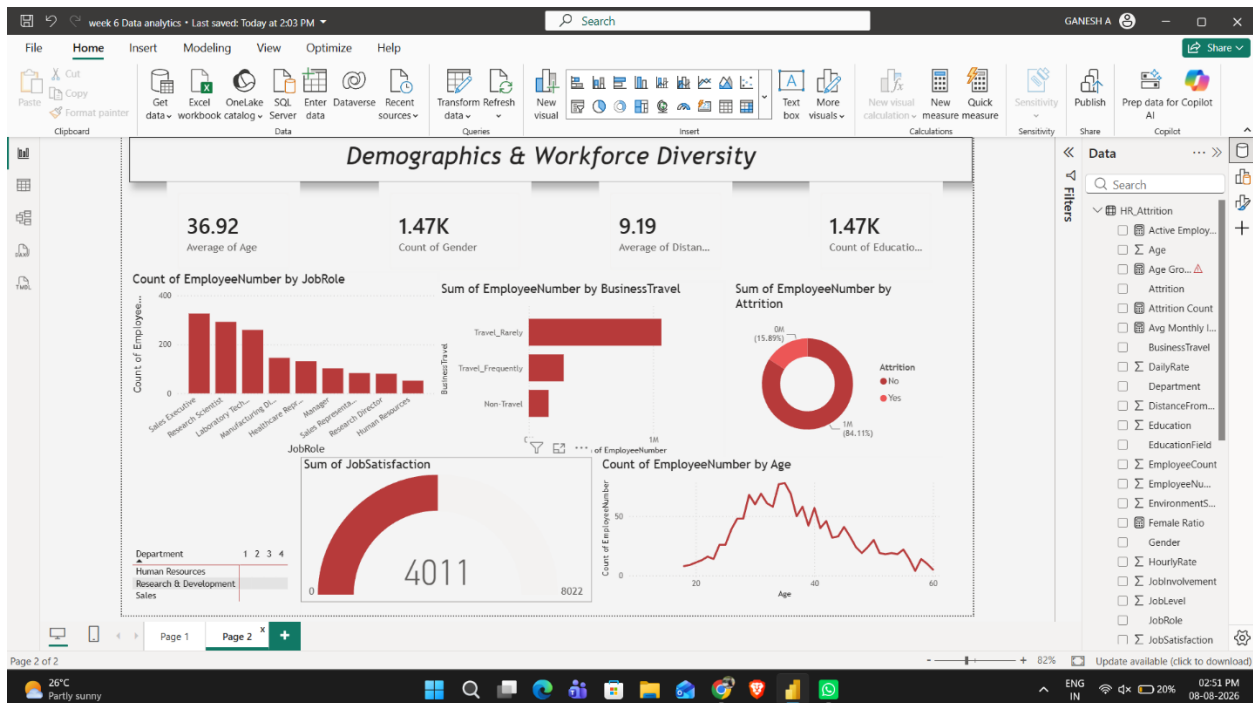
- Added Dax Queries to the KPI Cards

- Total Headcount = COUNTROWS('HR_Attrition').
- Active Employees = CALCULATE([Total Headcount], 'HR_Attrition'[Attrition] = "No").
- Attrition Count = CALCULATE([Total Headcount], 'HR_Attrition'[Attrition] = "Yes").
- Avg Monthly Income = AVERAGE('HR_Attrition'[MonthlyIncome]).



- Developed the Power BI interactive 2 page HR Analytics Dashboard





The dashboard enables the company to monitor Employee performance and attrition make data-driven decisions effectively And Improve Company Employees.

Result

- Successfully developed an interactive HR Attrition Analytics Dashboard using Power BI to support workforce planning and human capital management.
- Enabled HR leadership to monitor workforce metrics including headcount, active staff, attrition counts, and gender diversity ratios.
- Provided clear insights into department-wise and job role attrition as well as employee turnover across different tenure bands.
- Analyzed work environment impact on retention by examining overtime workload, business travel frequency, and commute distance.
- Explored employee satisfaction and age progression trends using line charts and gauge visuals to identify attrition risk triggers.
- Implemented interactive slicers for Department, Performance Rating, and Overtime to enable dynamic filtering and cross-analysis.
- Created a user-friendly executive report that supports data-driven talent management, retention strategies, and workforce planning.

Conclusion

The Walmart Sales Performance Dashboard was successfully developed using Power BI to transform sales data into meaningful and interactive visual reports. The dashboard provides a clear view of weekly sales performance, store-wise comparisons, holiday sales analysis, and sales trends over time. Interactive filters and visualizations make it easy for users to explore the data and gain valuable business insights. Overall, the dashboard supports effective decision-making by presenting sales information in a simple, organized, and user-friendly manner.

Learning Outcomes

Through this project, I learned:

- Gained practical experience in developing interactive HR dashboards using Power BI.
- Learned to create and customize different visualizations such as KPI cards, bar charts, donut charts, line graphs, matrix tables, and gauges.
- Understood how to analyze workforce data to identify attrition drivers, evaluate department-wise turnover, and measure employee satisfaction levels.
- Learned to use slicers and visual filters to enable dynamic cross-filtering and make dashboards interactive and user-friendly.
- Improved the ability to present complex demographic and talent metrics in a clean, visually appealing, and executive-ready format.
- Developed skills in converting raw HR data into meaningful workforce insights to support targeted employee retention and data-driven talent management.

Github Link:

https://github.com/GANESH-A08/Data_Analytics_projects/tree/main/Week6_Experiment